

question1 a)

```
In [1]: def max_3(a,b,c):  
        return max(a,b,c)  
result = max_3(10,25,15)  
print(result)
```

25

question 1 b)

```
In [5]: def even_odd(n):  
        if n % 2 == 0:  
            print(f"{n} is even")  
        else:  
            print(f"{n} is odd")  
even_odd(10)  
even_odd(7)
```

10 is even
7 is odd

question 1 c)

```
In [10]: def is_prime(n):  
        if n<=1:  
            return False  
        for i in range(2,n):  
            if n % i == 0:  
                return False  
        return True  
print(is_prime(5))  
print(is_prime(10))
```

True
False

question 1 d)

```
In [12]: def count_prime(n):  
        def is_prime(k):  
            if k<=1:  
                return False  
            for i in range(2,int(k**0.5)+1):  
                if k % i == 0:  
                    return False  
            return True  
  
        count=0  
        for num in range(2,n):  
            if is_prime(num):  
                count += 1  
        return count  
print(count_prime(20))
```

8

question 1 e)

```
In [14]: def rev_string(s):  
        reversed_str = ""  
        for char in s:  
            reversed_str=char+reversed_str  
        return reversed_str  
print(rev_string("world"))
```

dlrow

question 1 f)

```
In [15]: def rev_num(n):  
        reversed_num=0
```

```

negative=n<0
n=abs(n)

while n>0:
    digit=n%10
    reversed_num=reversed_num*10+digit
    n=n//10
return -reversed_num if negative else reversed_num
print(rev_num(25))
print(rev_num(-49))

```

52
-94

question 1 g)

```

In [16]: def fact(n):
        if n<0:
            return "factorial not defined for negative numbers"
        result=1
        for i in range(2,n+1):
            result *= i
        return result
print(fact(5))

```

120

question2

```

In [28]: def compute_pattern_sum(n):
        n1=int(str(n))
        n2=int(str(n)*2)
        n3=int(str(n)*3)
        return n1+n2+n3
print(compute_pattern_sum(5))

```

615

question3

```

In [18]: def count_vowels(s):
        vowels="aeiouAEIOU"
        count=0
        for char in s:
            if char in vowels:
                count += 1
        return count
print(count_vowels("spiderman"))

```

3

question 4

```

In [19]: def root_type(a,b,c):
        discriminant=b**2-4*a*c

        if discriminant>0:
            return "real and distinct roots"
        elif discriminant == 0:
            return "real and equal roots"
        else:
            return "complex roots"
print(root_type(1,5,6))
print(root_type(1,2,1))
print(root_type(1,2,5))

```

real and distinct roots
real and equal roots
complex roots

question 5

```

In [20]: def gcd(a,b):
        while b != 0:
            a, b=b, a%b

```

```
    return a
print(gcd(20,8))
```

4

question 6

```
In [21]: def lcm(a,b):
          def gcd(x,y):
              while y != 0:
                  x,y=y,x%y
              return x

          return abs(a*b)//gcd(a,b)
print(lcm(12,15))
```

60

question 8

```
In [24]: import math

def distance(x1,y1,x2,y2):
    return math.sqrt((x2-x1)**2+(y2-y1)**2)
print(distance(1,2,4,6))
```

5.0

question 9

```
In [25]: def is_perfect(n):
          if n<=0:
              return False

          sum_divisors=0
          for i in range(1,n):
              if n % i == 0:
                  sum_divisors += i

          return sum_divisors == n
print(is_perfect(6))
print(is_perfect(28))
print(is_perfect(12))
print(is_perfect(496))
print(is_perfect(8128))
```

True
True
False
True
True

question 10

```
In [27]: def fact_zeros(n):
          count=0
          i=5
          while n//i>0:
              count += n//i
              i *= 5
          return count
print(fact_zeros(11))
```

2

In []: